

EUROPEAN
SPALLATION
SOURCE

EasyDiffraction and crysta

Reusing crystallographic libraries, and building our own

Andrew Sazonov

European Spallation Source (ESS)
Data Management and Scientific Computing (DMSC)

LENS WG4 · Garching · 7 Oct 2026

1 WHY NEW SOFTWARE

2 EASYDIFFRACTION

3 CRYSTA

4 STATUS AND PLANS

1 WHY NEW SOFTWARE
2 EASYDIFFRACTION
3 CRYSTA
4 STATUS AND PLANS

1.1 WHY NEW SOFTWARE

Diffraction analysis: many tools

Many tools, each with its own interface.

DIFFRACTION PROGRAMS

FullProf Suite · Jana2020 · GSAS-II · SHELX · MAUD · FOX · Mag2Pol · TOPAS · HighScore · PDFgui · …

DIFFRACTION LIBRARIES

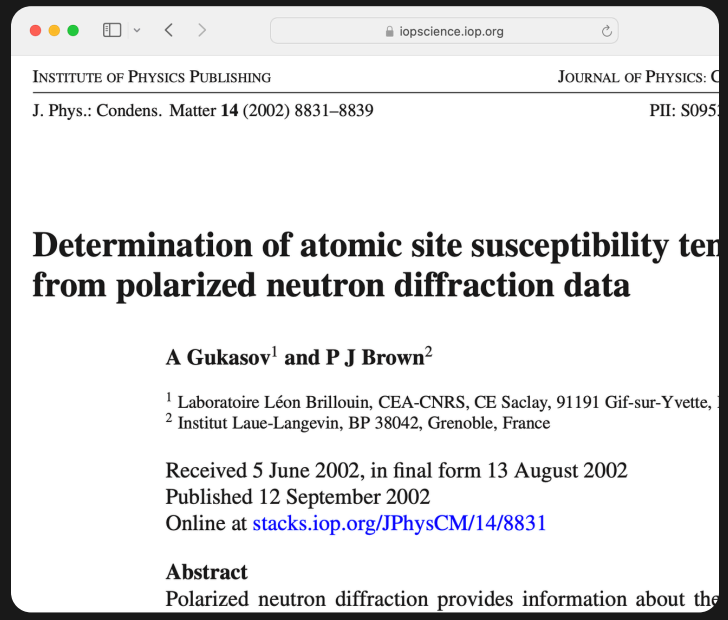
CrysFML · CCSL · cctbx · pyobjcryst · CrysPy · PDFfit2 · GSAS-II · …

1.2 WHY NEW SOFTWARE

What users run into

Site susceptibility

CrysPy, CCSL only · for MAGIC



iopscience.iop.org

INSTITUTE OF PHYSICS PUBLISHING — JOURNAL OF PHYSICS: C

J. Phys.: Condens. Matter **14** (2002) 8831–8839 — PII: S095

Determination of atomic site susceptibility ter from polarized neutron diffraction data

A Gukasov[1] **and P J Brown**[2]

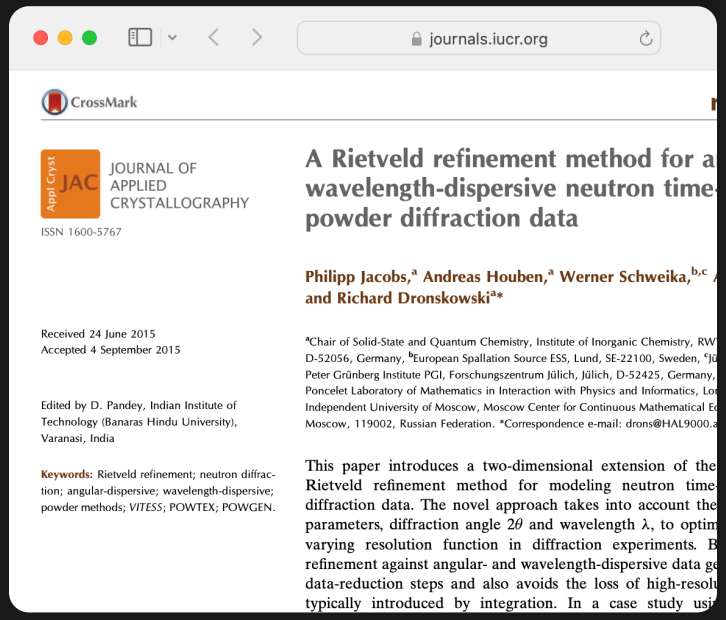
[1] Laboratoire Léon Brillouin, CEA-CNRS, CE Saclay, 91191 Gif-sur-Yvette,
[2] Institut Laue-Langevin, BP 38042, Grenoble, France

Received 5 June 2002, in final form 13 August 2002
Published 12 September 2002
Online at stacks.iop.org/JPhysCM/14/8831

Abstract
Polarized neutron diffraction provides information about the

2D Rietveld

modified GSAS-II only · for DREAM



journals.iucr.org

CrossMark

Appl Cryst JAC JOURNAL OF APPLIED CRYSTALLOGRAPHY

ISSN 1600-5767

A Rietveld refinement method for a wavelength-dispersive neutron time- powder diffraction data

Philipp Jacobs,[a] **Andreas Houben,**[a] **Werner Schweika,**[b,c] **and Richard Dronskowski**[a]*

[a]Chair of Solid-State and Quantum Chemistry, Institute of Inorganic Chemistry, RW D-52056, Germany, [b]European Spallation Source ESS, Lund, SE-22100, Sweden, [c]Jü Peter Grünberg Institute PGI, Forschungszentrum Jülich, Jülich, D-52425, Germany, Poncelet Laboratory of Mathematics in Interaction with Physics and Informatics, Lo Independent University of Moscow, Moscow Center for Continuous Mathematical Ed Moscow, 119002, Russian Federation. *Correspondence e-mail: drons@HAL9000.a

Received 24 June 2015
Accepted 4 September 2015

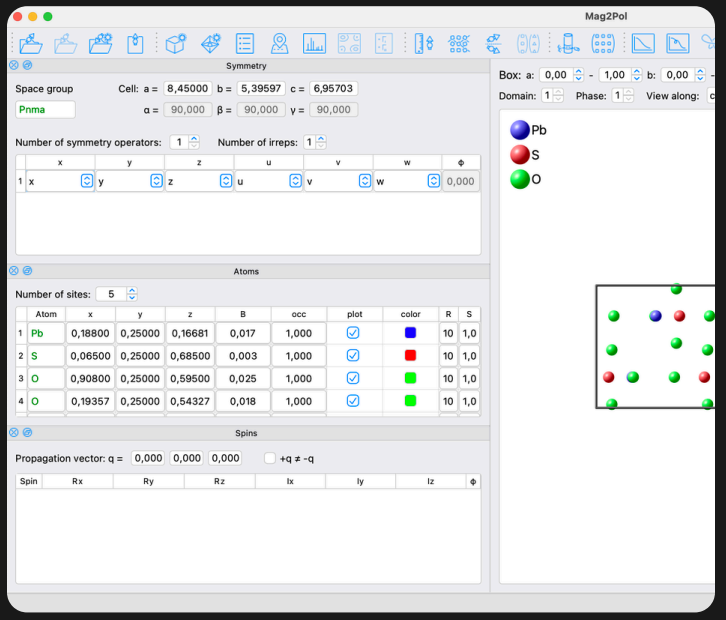
Edited by D. Pandey, Indian Institute of Technology (Banaras Hindu University), Varanasi, India

Keywords: Rietveld refinement; neutron diffraction; angular-dispersive; wavelength-dispersive; powder methods; *VITESS*; POWTEX; POWGEN.

This paper introduces a two-dimensional extension of the Rietveld refinement method for modeling neutron time- diffraction data. The novel approach takes into account the parameters, diffraction angle 2θ and wavelength λ, to optim varying resolution function in diffraction experiments. B refinement against angular- and wavelength-dispersive data ge data-reduction steps and also avoids the loss of high-resolu typically introduced by integration. In a case study usi

Not the best UX

many panels and controls at once



What users run into

Gaps for the ESS instruments, and a steep start for new users.

High entrance barrier

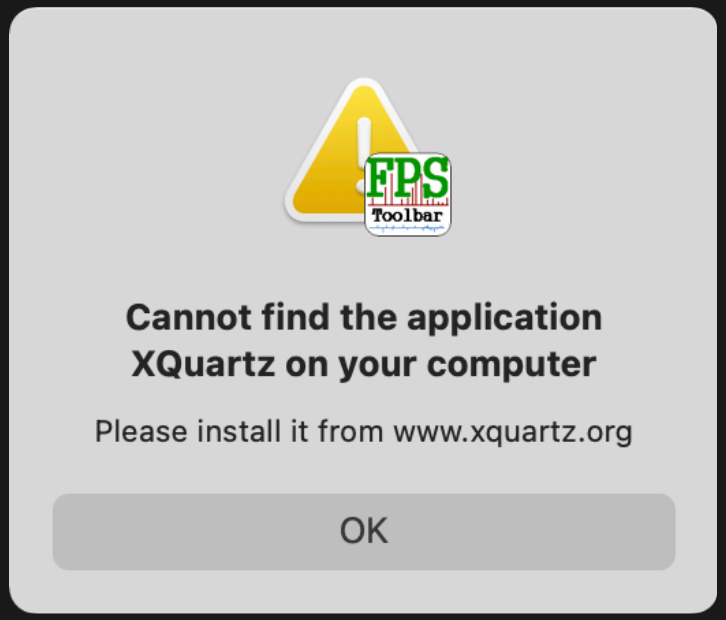
input files with cryptic labels

cecual.pcr

```
COMM CeCuAl3 RT #13744 ThCr2Si2 type
! Current global Chi2 (Bragg contrib.) =      2.273
! Files => DAT-file: cecual.dat,  PCR-file: cecual
!Job Npr Nph Nba Nex Nsc Nor Dum Iwg Ilo Ias Res Ste Nre Cry Uni Cor Opt Aut
  -1   9   1  -4   2   0   0   0   0   0   0   0   0   0   1   0   0   1
!
!Ipr Ppl Ioc Mat Pcr Ls1 Ls2 Ls3 NLI Prf Ins Rpa Sym Hkl Fou Sho Ana
   0   0   1   0   1   0   4   0   0  -3  10  -1   1   0   0   0   0
!
!  Bkpos        Wdt    Iabscor for Pattern#  1
    7000.000    3.20     2
!NCY  Eps  R_at  R_an  R_pr  R_gl    TOF-min      <Step>       TOF-max
 10  0.10  1.00  1.00  1.00  1.00    3001.5891      4.3357  19004.6582
!
! Excluded regions (LowT  HighT) for Pattern#  1
    2000.10     3000.00
   19000.00    20000.00
!
!
     30    !Number of refined parameters
!
!    Zero    Code      Dtt1    Code      Dtt2    Code   2ThetaBank  -> Patt# 1
    2.921  131.00  6167.247    0.00    -2.280    0.00   145.000
!
!   Background coefficients/codes  for Pattern#  1  (Fourier cosine series, up to 18 coefficients)
   24832.850    6139.245    8063.473    3125.051    2566.957     311.078
      31.00       41.00       51.00       91.00       101.00       111.00
     837.348    -103.742    -11.806       0.000       0.000        0.000
     271.00      231.00      241.00        0.00        0.00         0.00
      0.000       0.000       0.000       0.000       0.000        0.000
       0.00        0.00        0.00        0.00        0.00         0.00
!-------------------------------------------------------------------------------
!  Data for PHASE number:   1  ==> Current R_Bragg for Pattern#  1:   2.2753
!-------------------------------------------------------------------------------
CeCuAl3
!
!Nat Dis Ang Pr1 Pr2 Pr3 Jbt Irf Isy Str Furth       ATZ    Nvk Npr More
   6   0   0 0.0 0.0 1.0   0   0   0   0   0      36190.043   0   9   1
!
!Jvi Jdi Hel Sol Mom Ter  Brind   RMua    RMub    RMuc   Jtyp  Nsp_Ref Ph_Shift N_Domains
   0   0   0   0   0   0  1.0000  0.0000  0.0000  0.0000   -1      0       0        0
```

Extra dependencies

XQuartz to install on macOS



Not cross-platform

no macOS or Linux support

	Download from
Jana2020 for Windows	jana2020Inst.msi
Jana2006 for Windows, STABLE	janainst.msi
Jana2006 for Unix, STABLE	not available
Jana2006 for Windows, LATEST	janainst.msi
Jana2006 for Unix, LATEST	not available

1.3 WHY NEW SOFTWARE

Needs

No existing program offers all three.

- Extend functionality for the ESS instruments
- Improve usability for every user
- Simplify maintenance for a small team

1 WHY NEW SOFTWARE

2 EASYDIFFRACTION

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2.1 EASYDIFFRACTION

Built on existing libraries

We start from the libraries, not the programs.

DIFFRACTION PROGRAMS

FullProf Suite · Jana2020 · GSAS-II · SHELX · MAUD · FOX · Mag2Pol · TOPAS · HighScore · PDFgui · …

DIFFRACTION LIBRARIES

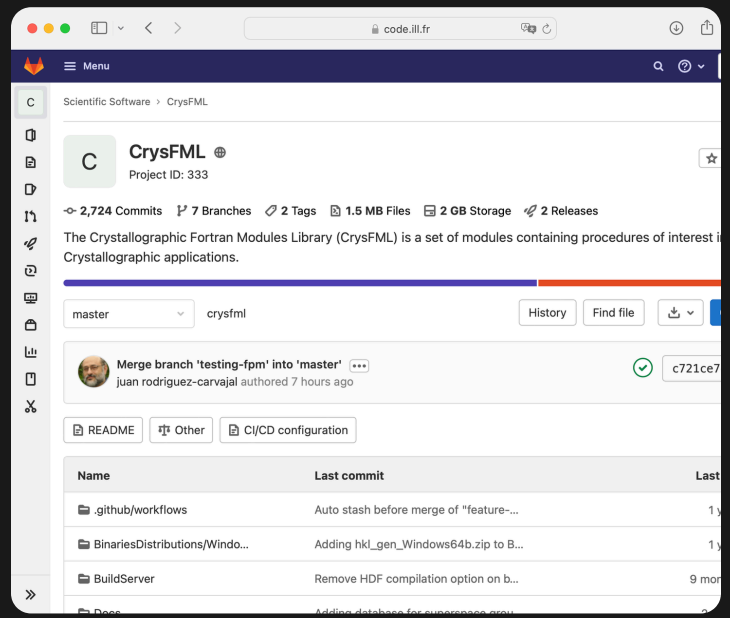
CrysFML · CCSL · cctbx · pyobjcryst · CrysPy · PDFfit2 · GSAS-II · …

What each library brings

Each for the functionality we need.

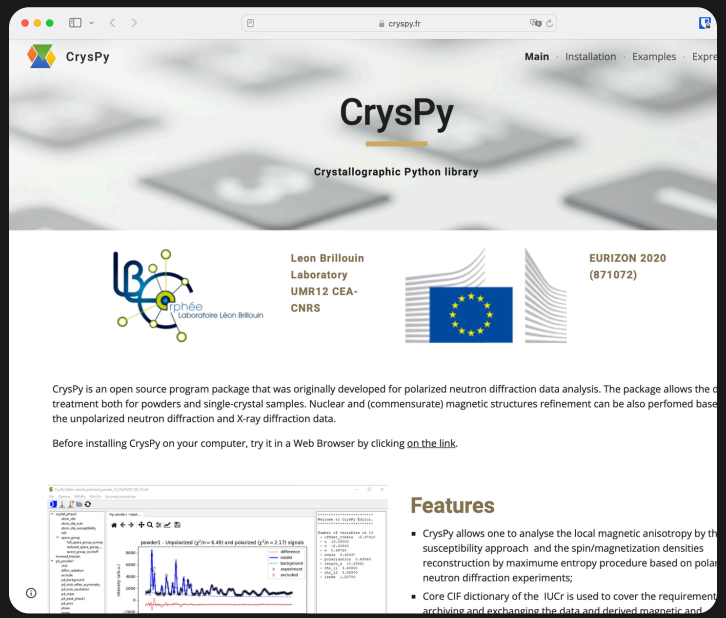
CrysFML

mature, wide coverage · Fortran



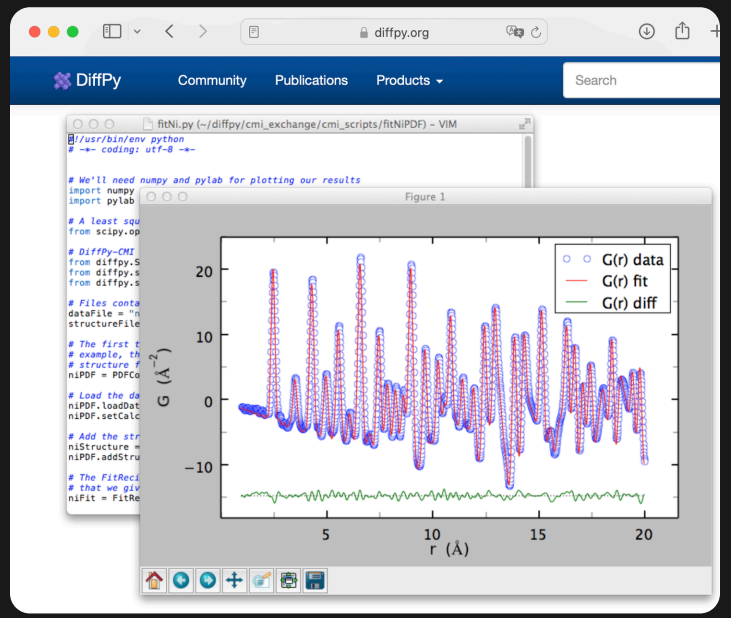
CrysPy

more polarised-neutron options · Python



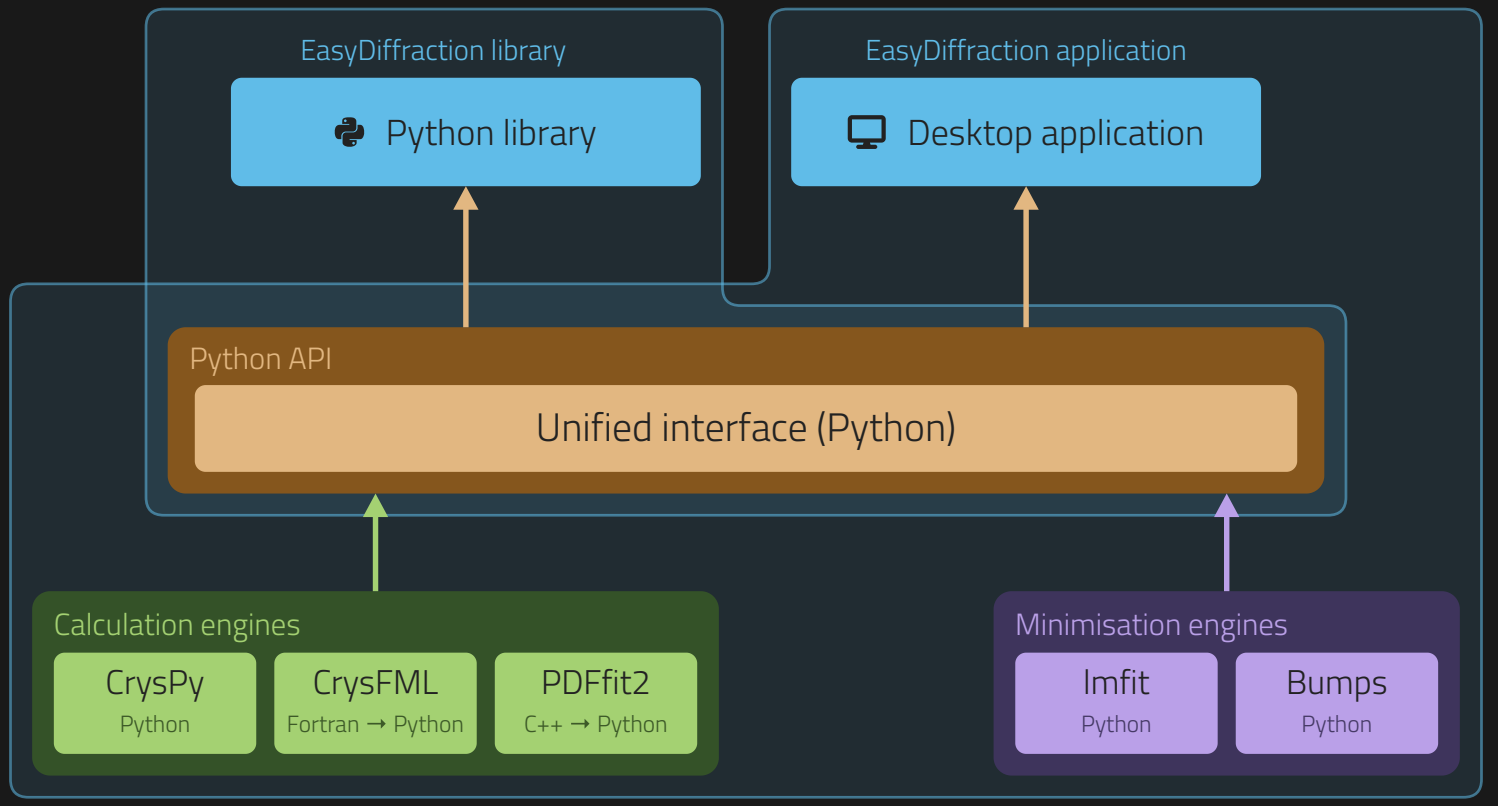
PDFfit2

pair distribution function · C++



What we rely on

Calculations come from the libraries; the interface is ours.



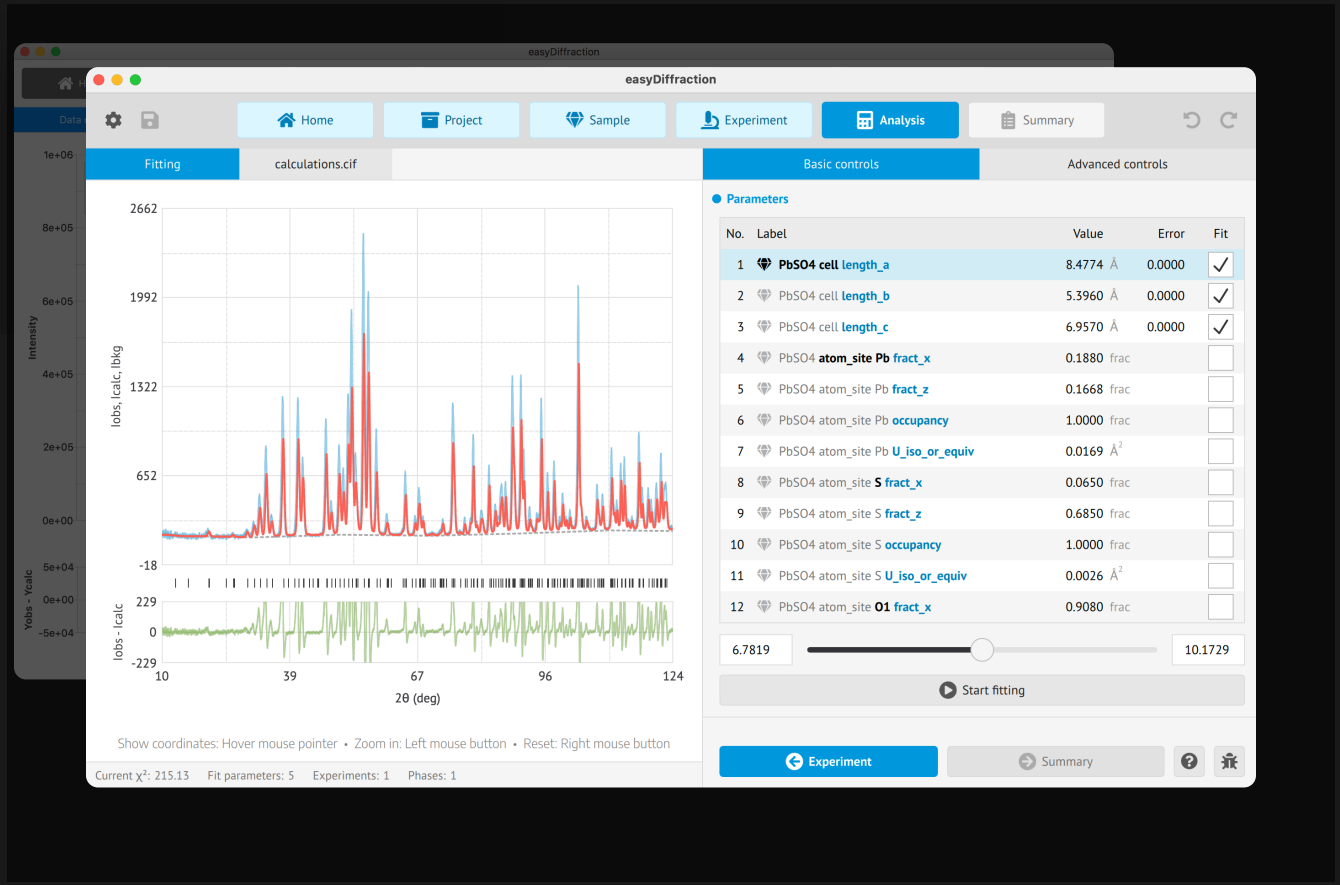
Several years of improving UI/UX

2019 · the first mock-up, for diffraction



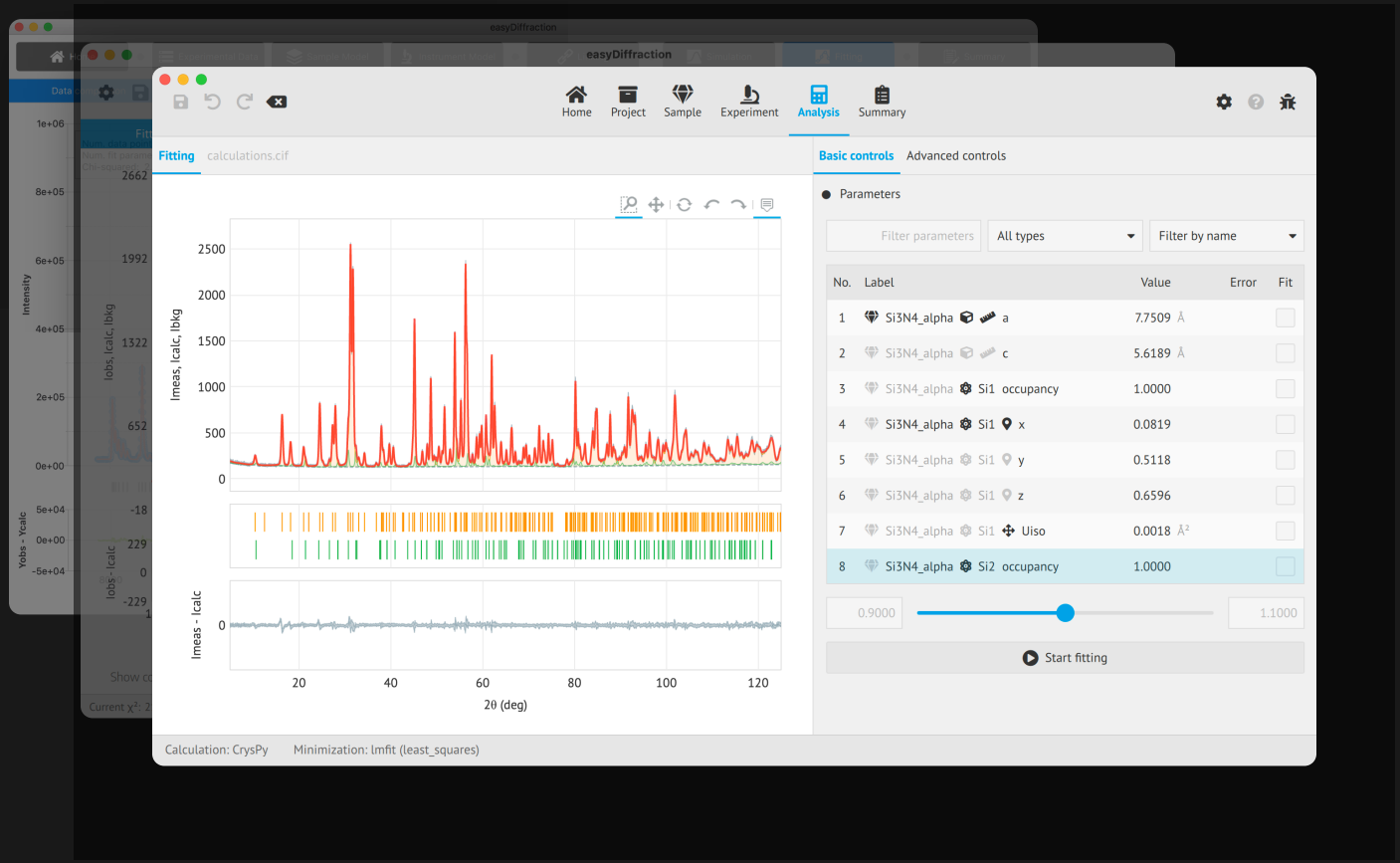
Several years of improving UI/UX

2020 · a simpler workflow



Several years of improving UI/UX

2021 · cleaner, lighter



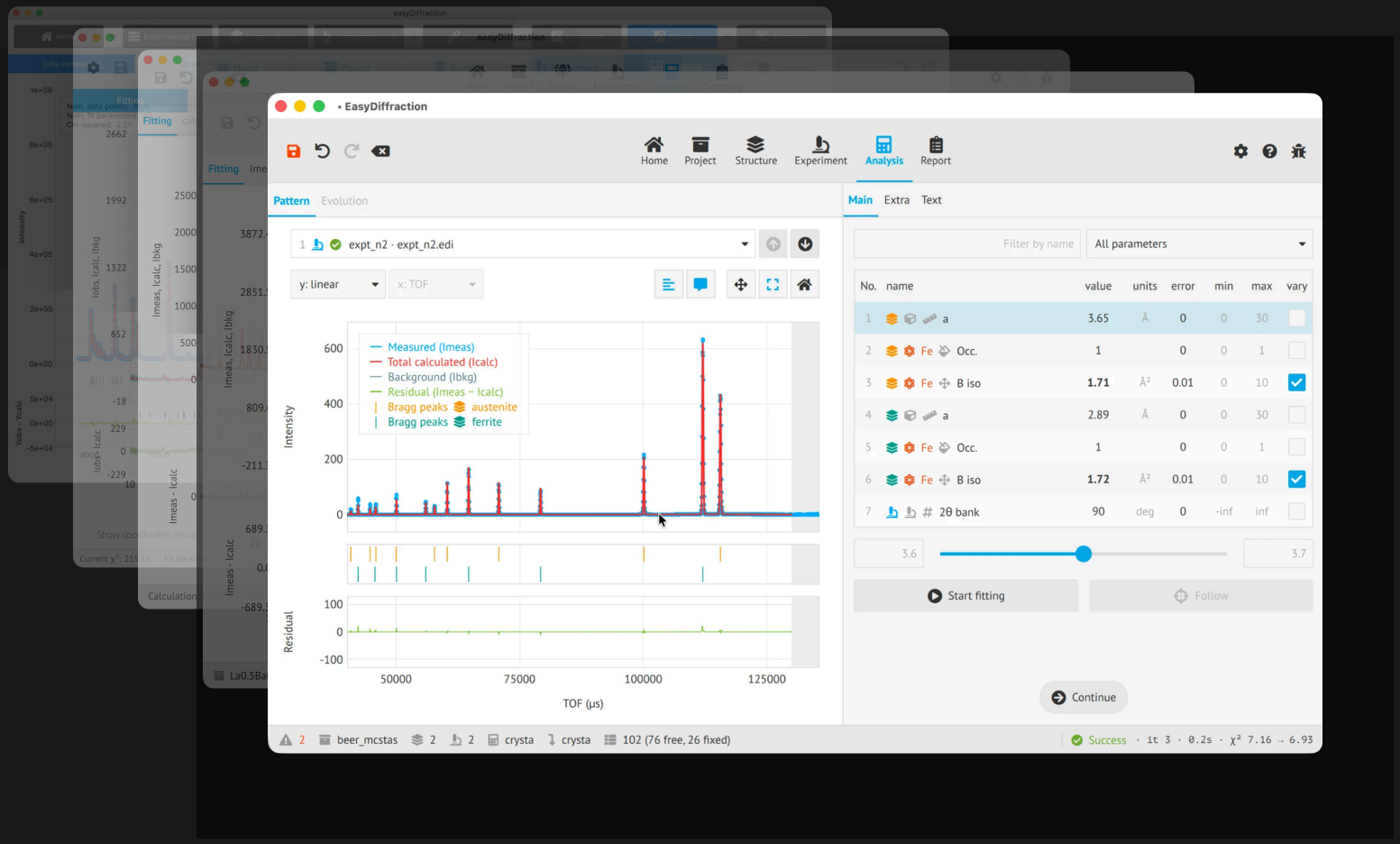
Several years of improving UI/UX

2023–2025 · the same shape, easier to use



Several years of improving UI/UX

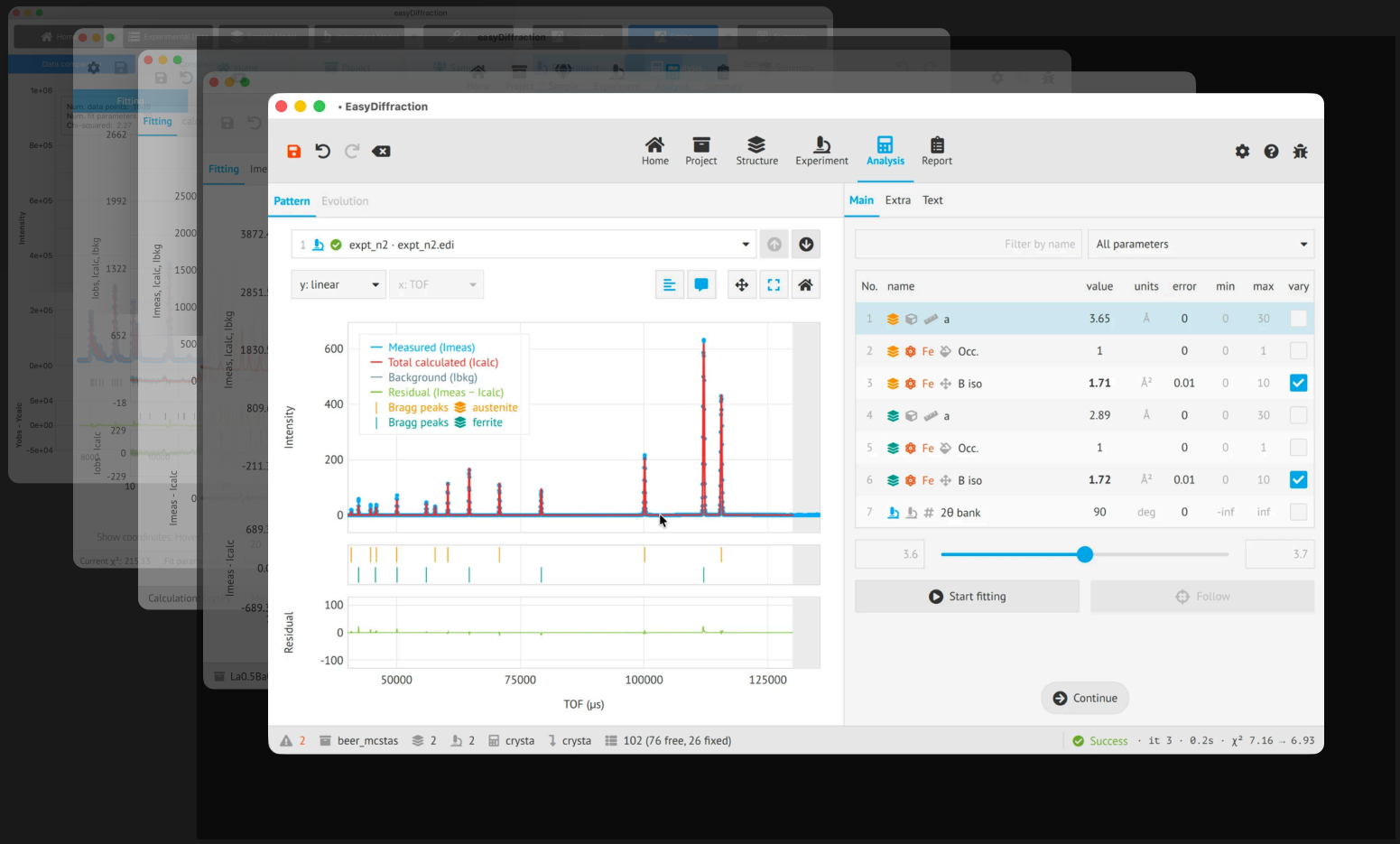
2026 · more functionality, still intuitive



Several years of improving UI/UX

The functionality grew; the user interface stayed simple.

2026 · more functionality, still intuitive

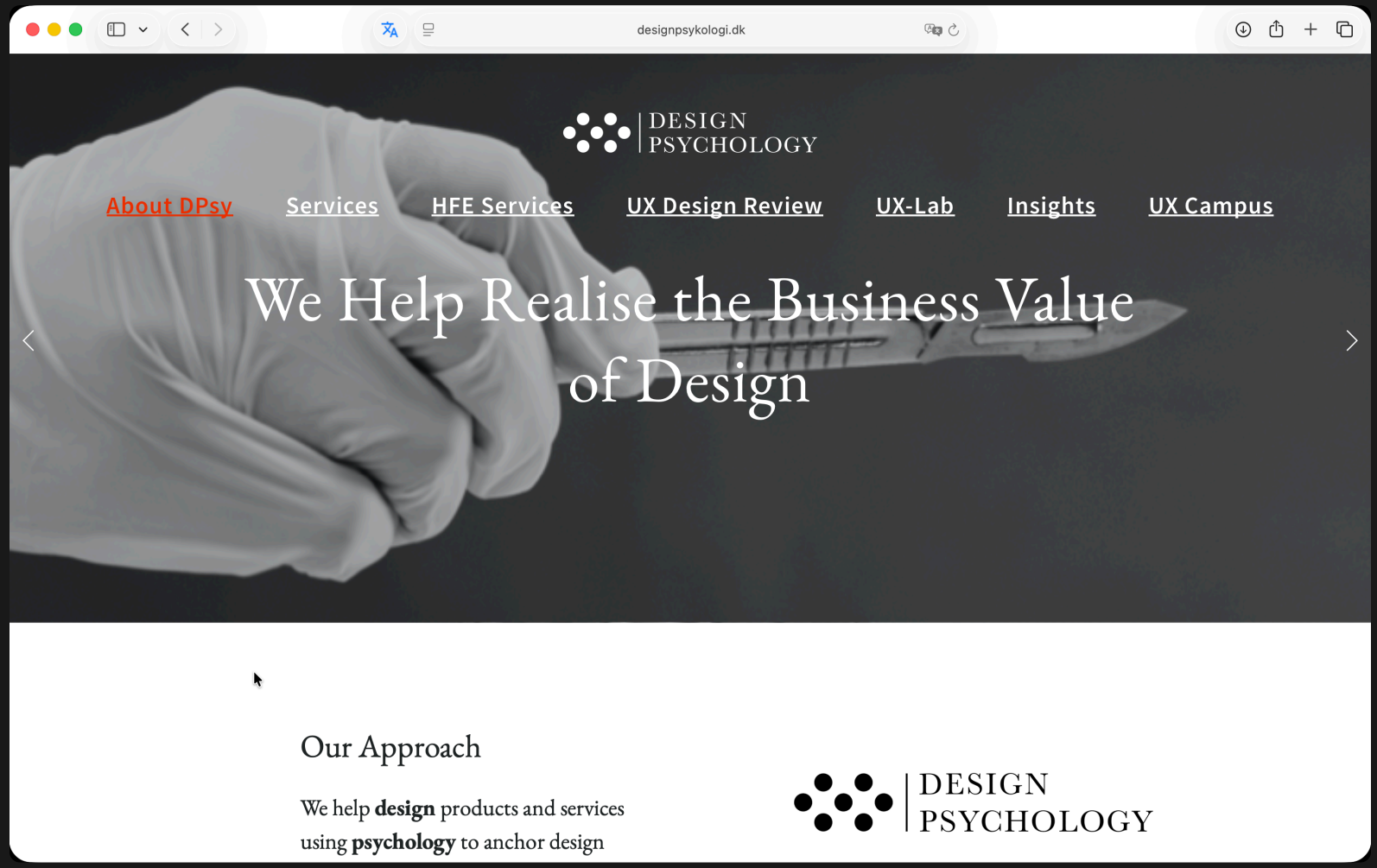


2.5 EASYDIFFRACTION

Reviewed by UX experts

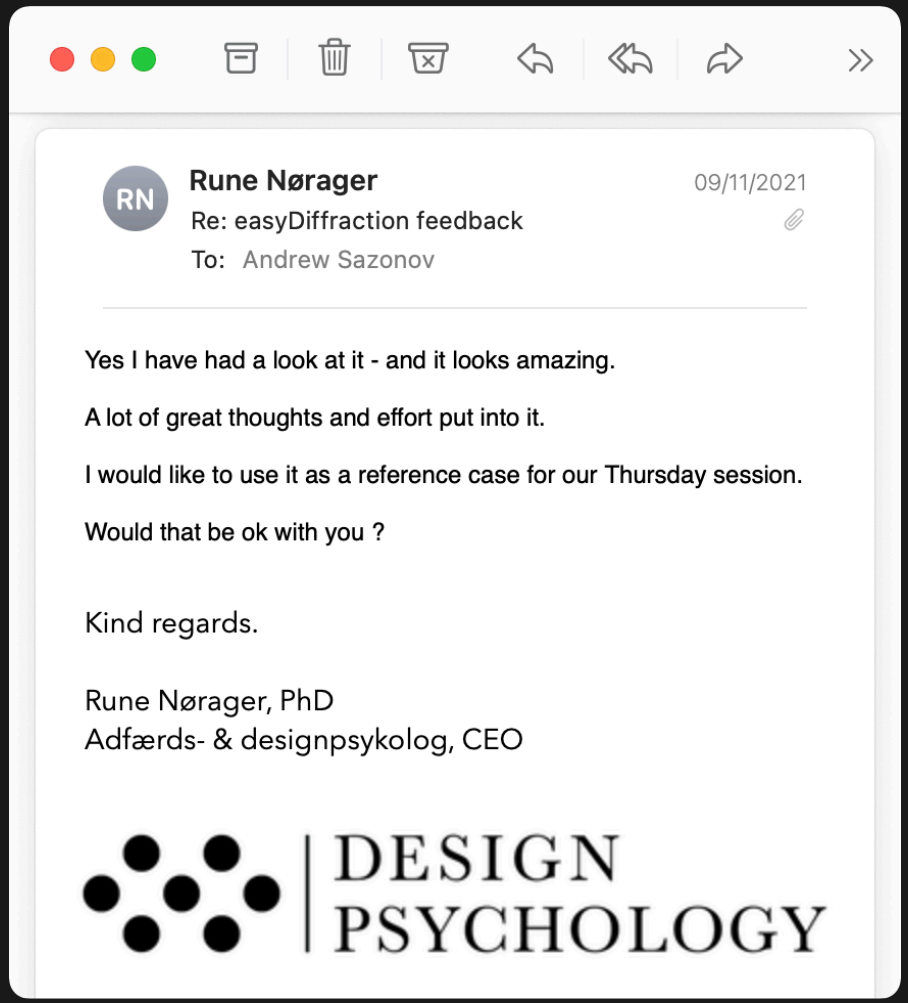
Reviewed by UX experts

Two workshops with Design Psychology, a UX consultancy.



Reviewed by UX experts

Then they reviewed our interface.



Rune Nørager 09/11/2021
Re: easyDiffraction feedback
To: Andrew Sazonov

Yes I have had a look at it - and it looks amazing.

A lot of great thoughts and effort put into it.

I would like to use it as a reference case for our Thursday session.

Would that be ok with you ?

Kind regards.

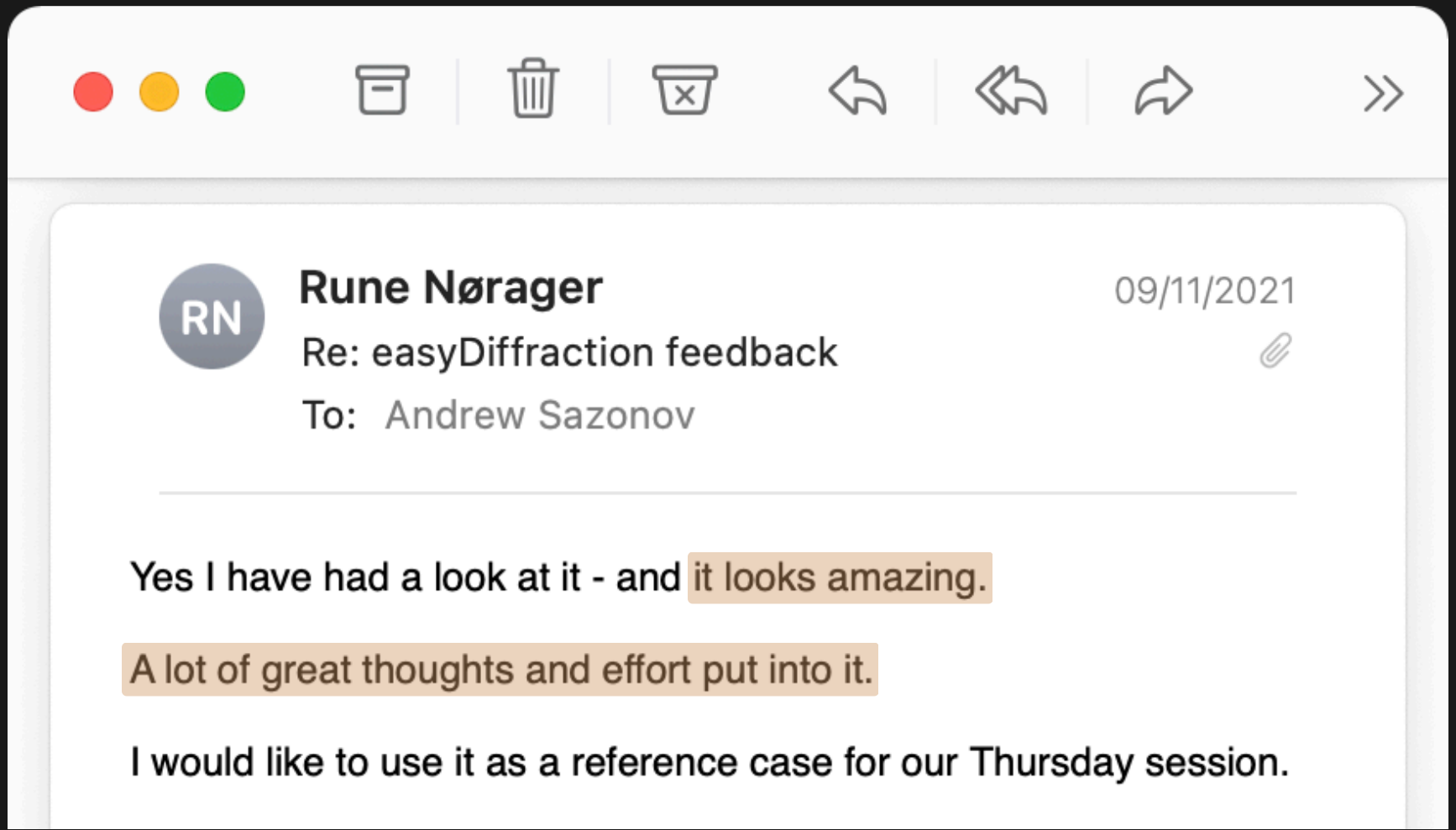
Rune Nørager, PhD
Adfærds- & designpsykolog, CEO

DESIGN PSYCHOLOGY

Reviewed by UX experts

Their verdict: a well-designed interface.

Then they reviewed our interface.

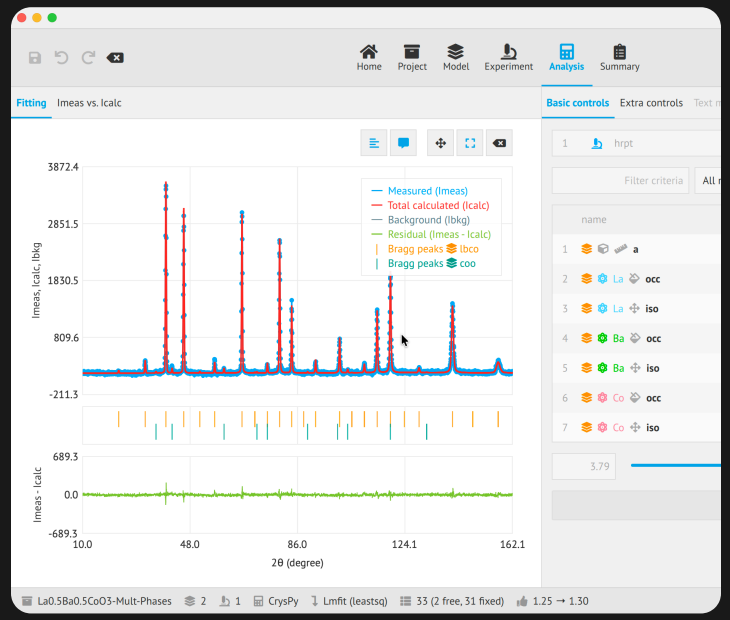


Three ways to run the same workflow

The same project, whichever way you open it.

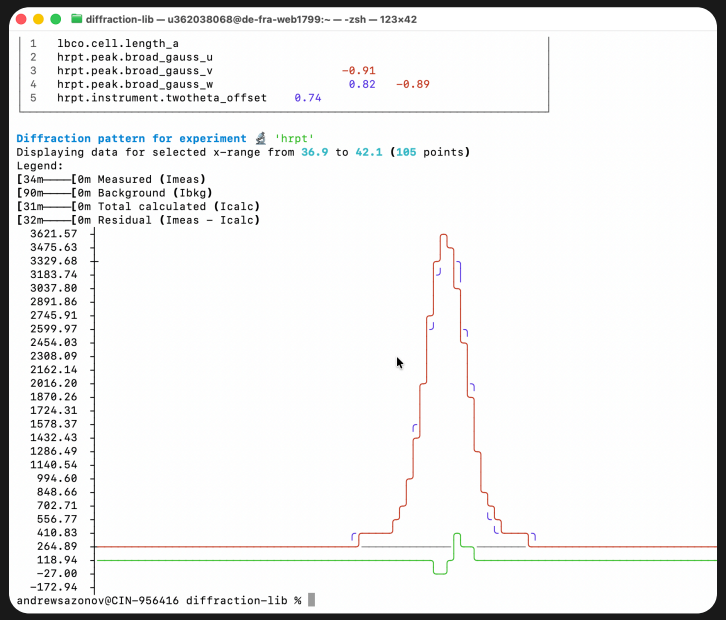
Graphical interface

for users who prefer to click



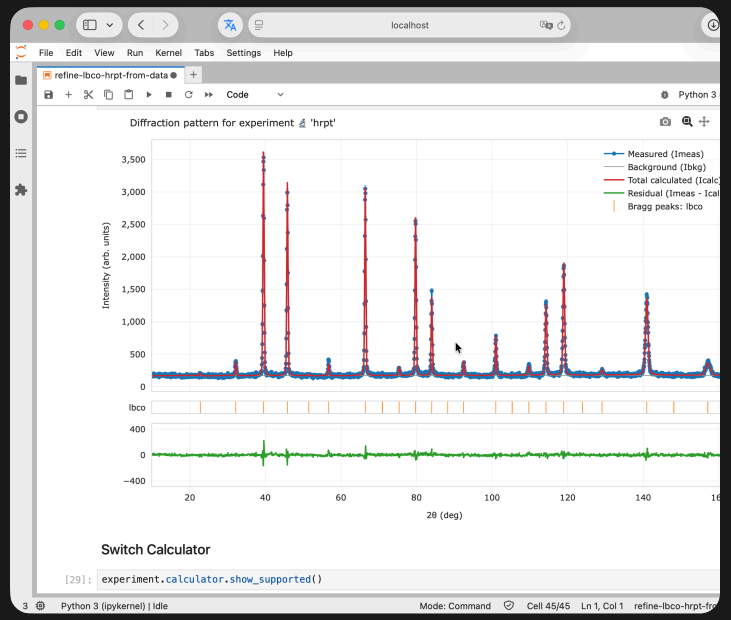
Command line

for command-line enthusiasts



Python library

for those who prefer scripts/Jupyter



Same look and feel, different technique

The same layout works for other techniques, even without fitting.

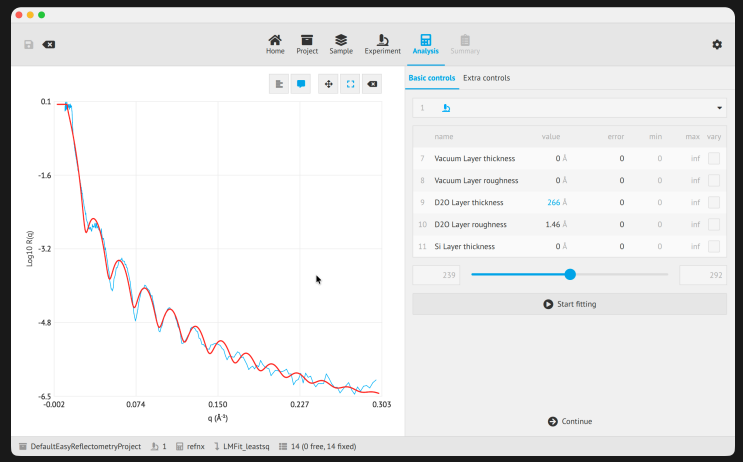
EasyDiffraction

a powder pattern and a crystal structure



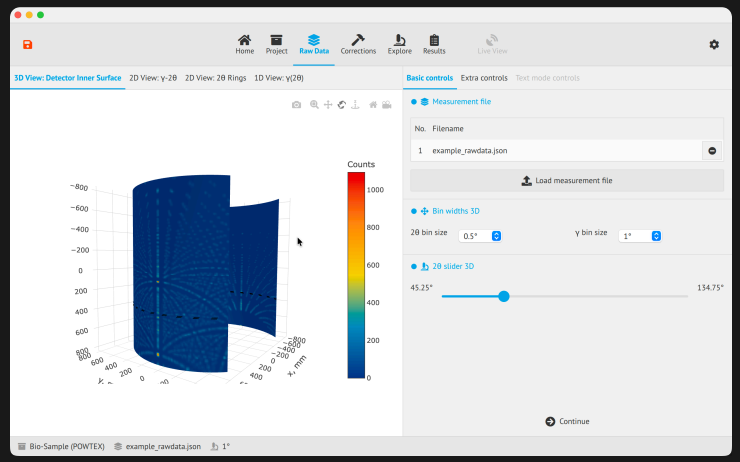
EasyReflectometry

a reflectivity curve and a layer stack



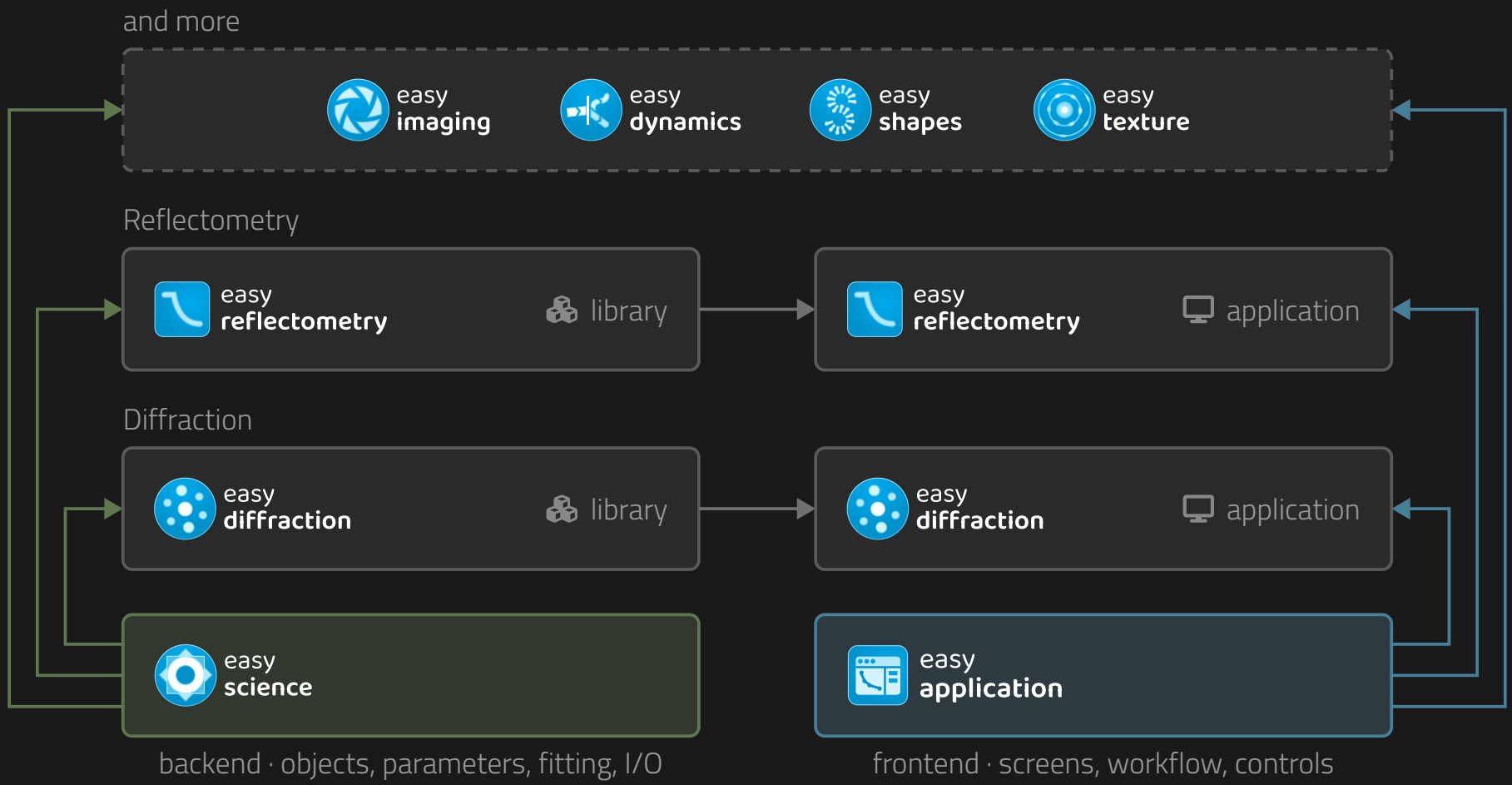
EasyTexture

raw detector data, before any model



What became reusable

With EasyScience, a new technique adds only its science.



Where EasyScience is used today

Projects at different stages, some with partners.

easyscience.org · github.com/easyscience

easy diffraction
analysis · running
diffraction

easy reflectometry
analysis · running
reflectometry

easy dynamics
analysis · new
quasielastic neutron scattering

easy imaging
analysis · new
Bragg-edge imaging

easy texture
reduction · external
FZ Jülich · Göttingen
texture

easy shapes
modelling · new
Shapespyre · Chalmers
molecular structures

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3.1 CRYSTA

Back to diffraction: what analysis needs

EasyDiffraction works today, but not fast enough.

- Large series · 1,000 datasets × 100 patterns · temperature, field, time
- Multidimensional data · huge parameter spaces · 2D and 3D detectors
- Automation · experiments that run themselves · no human in the loop

3.2 CRYSTA

AI in crystallography

AI is now widely used, from coding to experiment orchestration.



27th Congress and General Assembly
of the International Union of Crystallography

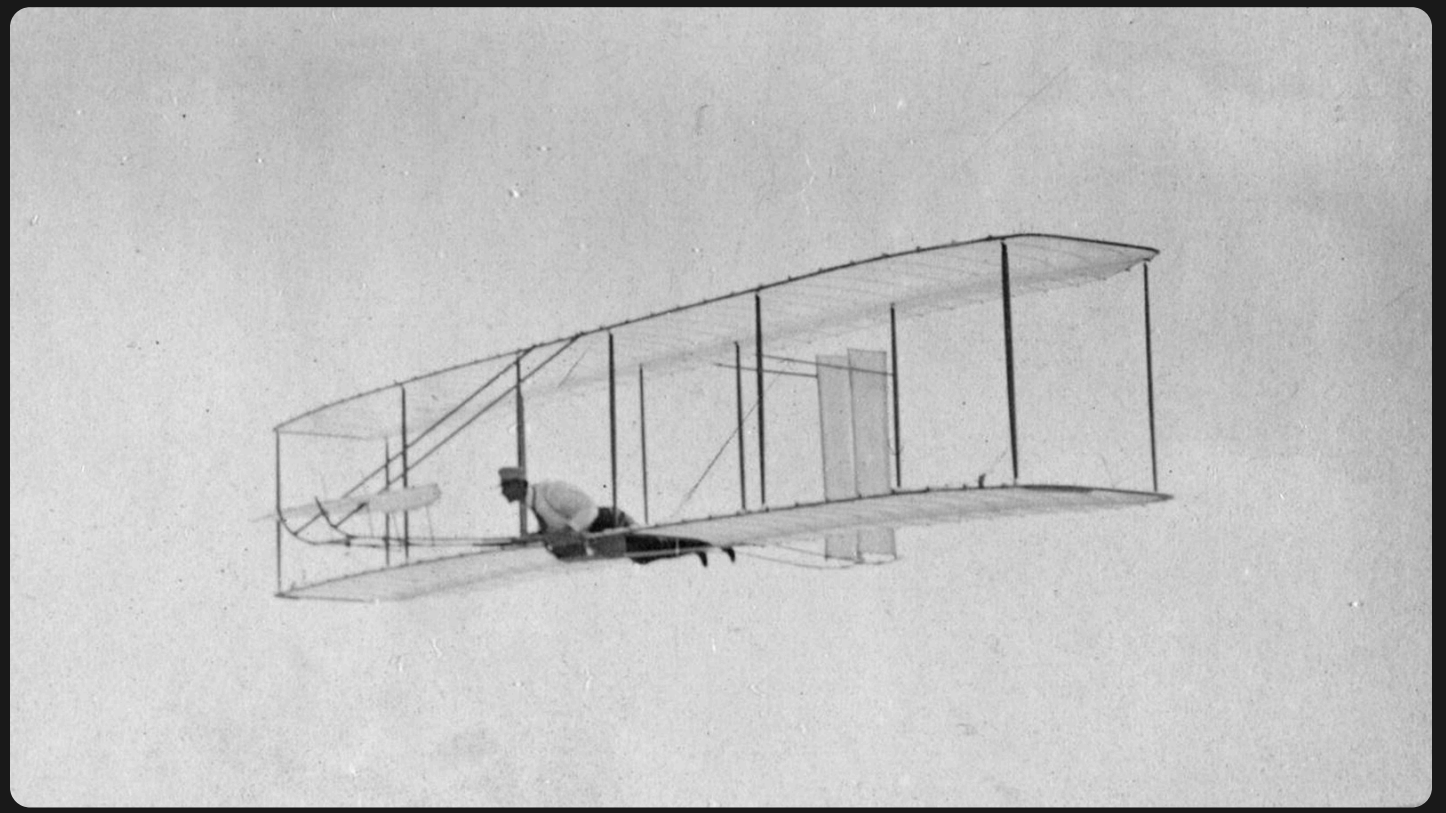
Calgary, August 2026 · ≈ 1,800 attendees

AI or machine learning named in the title or abstract

23	121	69
of 142 microsymposia	of 839 microsymposium talks	of 932 posters

A pilot project

Can supervised AI build a fast crystallographic library?



3.4 CRYSTA

Meet crysta

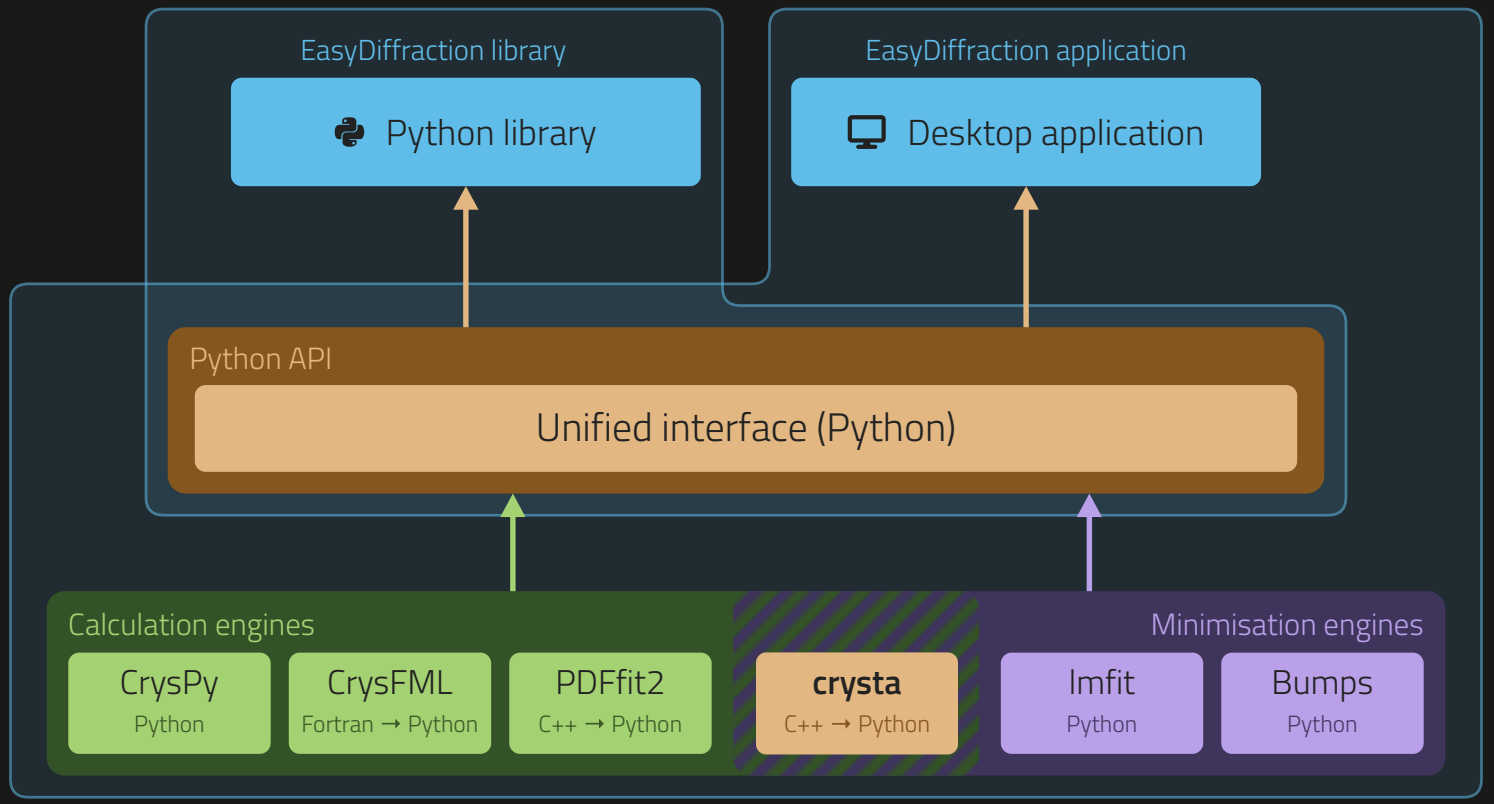
crysta

Eigen · nanobind · CMake



Meet crysta

A new calculation and minimisation engine, built in-house.



The test case

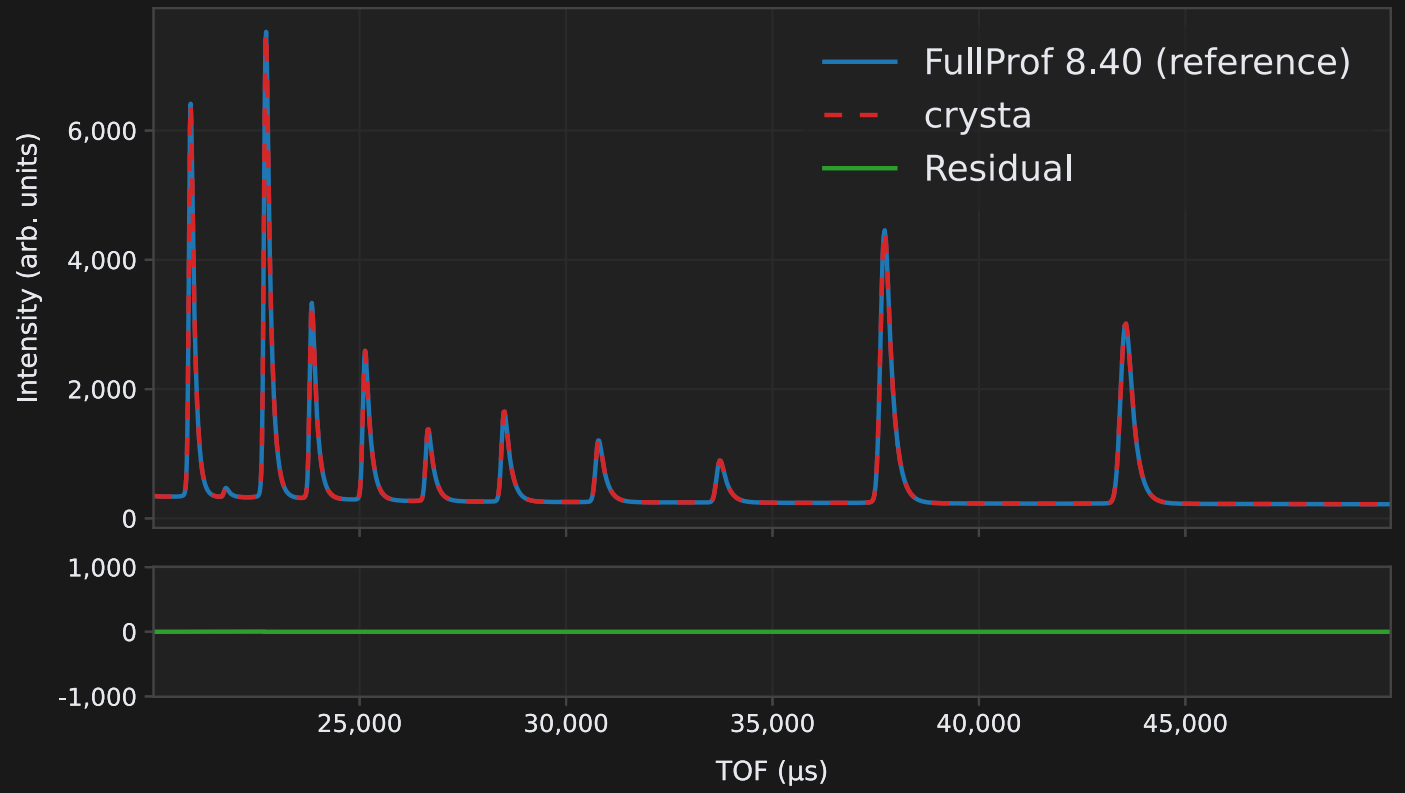
One real case, the same for every code.

- **Sample** · NCAF, $Na_2Ca_3Al_2F_{14}$ · reference powder, cubic
- **Measurement** · WISH at ISIS, banks 2 + 9 · 3977 points, 627 peaks
- **Refinement** · 48 free parameters · structure, profile, background
- **Method** · weighted least squares · same start, $\chi^2 \approx 4.8$ for all
- **Reference** · FullProf 8.40 · same model, one thread

Agreement of calculated patterns

Largest difference 3.9 of 7527 counts: 0.05 %.

NCAF | WISH 2+9 | 3977 points | 627 peaks



Agreement of refined parameters

Every parameter within 0.69 σ of FullProf, median 0.05 σ.

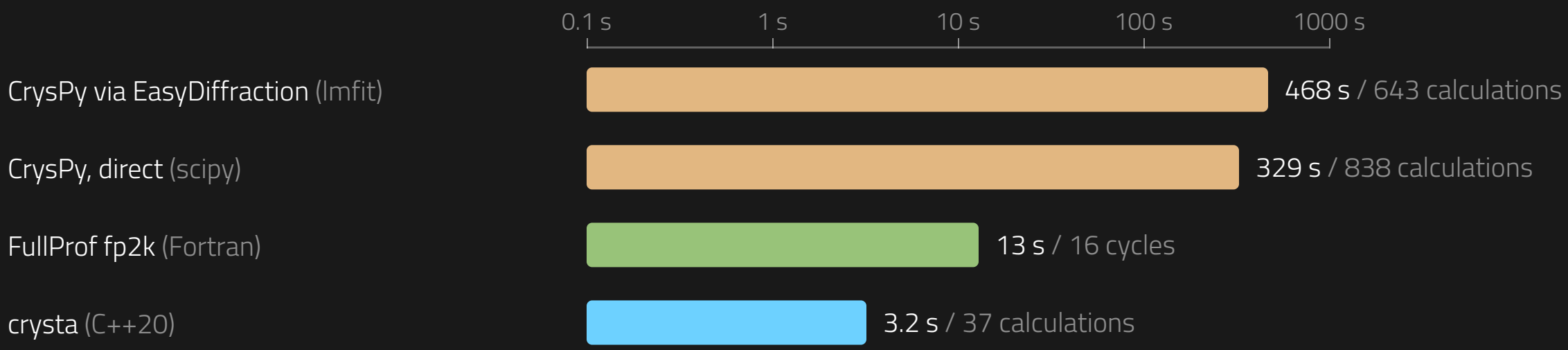
NCAF | WISH 2+9 | 3977 points | 627 peaks | same start | $\chi^2 \approx 4.8$ | 48 parameters

parameter	crysta	FullProf	Δ	Δ/σ
background[21]	164 ± 22	187 ± 25	−23	0.69
β_0 (decay)	0.00516 ± 0.00004	0.00518 ± 0.00004	−0.000023	0.39
… 44 more …				
Ca1 B_{iso}	1.12 ± 0.13	1.12 ± 0.13	−0.0017	0.01
F3 B_{iso}	1.85 ± 0.16	1.85 ± 0.17	−0.0004	0.00

Speed: a full refinement

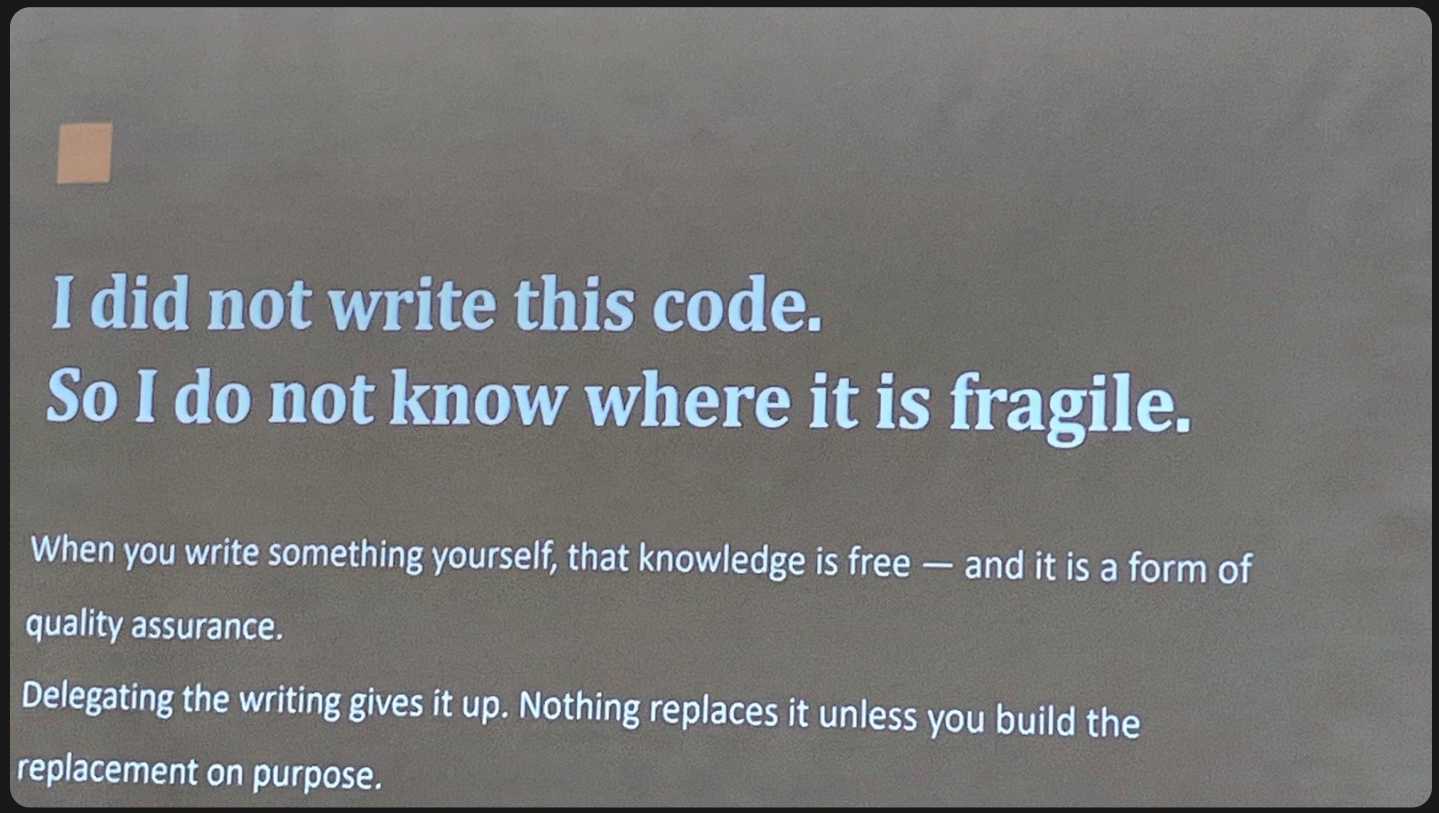
150× faster than today's EasyDiffraction route.

NCAF | WISH 2+9 | 3977 points | 627 peaks | 1 thread | log scale



A lesson from IUCr 2026

Nobody knows where AI-written code is fragile until it is tested.



Oleg Sobolev & Paul Adams, LBNL · AI-written CIF parser for cctbx · MS-116, IUCr 2026, Calgary

How the code is checked

AI writes the code, so it is checked more, not less.

- **Tests** at every level · unit, integration, system
- **Independent references** · FullProf, CrysFML, CrysPy
- **Strict builds** · warnings are errors, static analysis
- **Continuous integration** · every change, Linux, macOS and Windows

3.11 CRYSTA

Credits

Decades of work by many people, most of it long before AI.

LITERATURE

International Tables A and C · Cromer & Mann · Waasmaier & Kirfel · Busing & Levy · Caglioti · Thompson, Cox & Hastings · Finger, Cox & Jephcoat · Von Dreele · Ikeda & Carpenter · Becker & Coppens · Dollase · Rietveld · Marquardt · Proffen & Billinge · Egami & Billinge · Blume · Bertaut · Wills · Sears · Lovesey · Shmueli · Prince · Larson & Von Dreele · …

OPEN-SOURCE SOFTWARE

CrysPy: Iurii Kibalin (ESS), Arsen Gukasov (ILL), Andrew Sazonov (ESS) · **CrysFML**: Juan Rodríguez-Carvajal (ILL), Javier González-Platas (ULL), Nebil Ayape Katcho (ILL) · **cctbx**: R. Grosse-Kunstleve, P. Adams, N. Sauter et al. (LBNL) · **GSAS-II**: B. Toby, R. Von Dreele (ANL) · **PDFfit2**: C. Farrow, P. Juhás, T. Proffen, S. Billinge et al. · …

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How much is done

crysta after eight weeks: how far can it go?

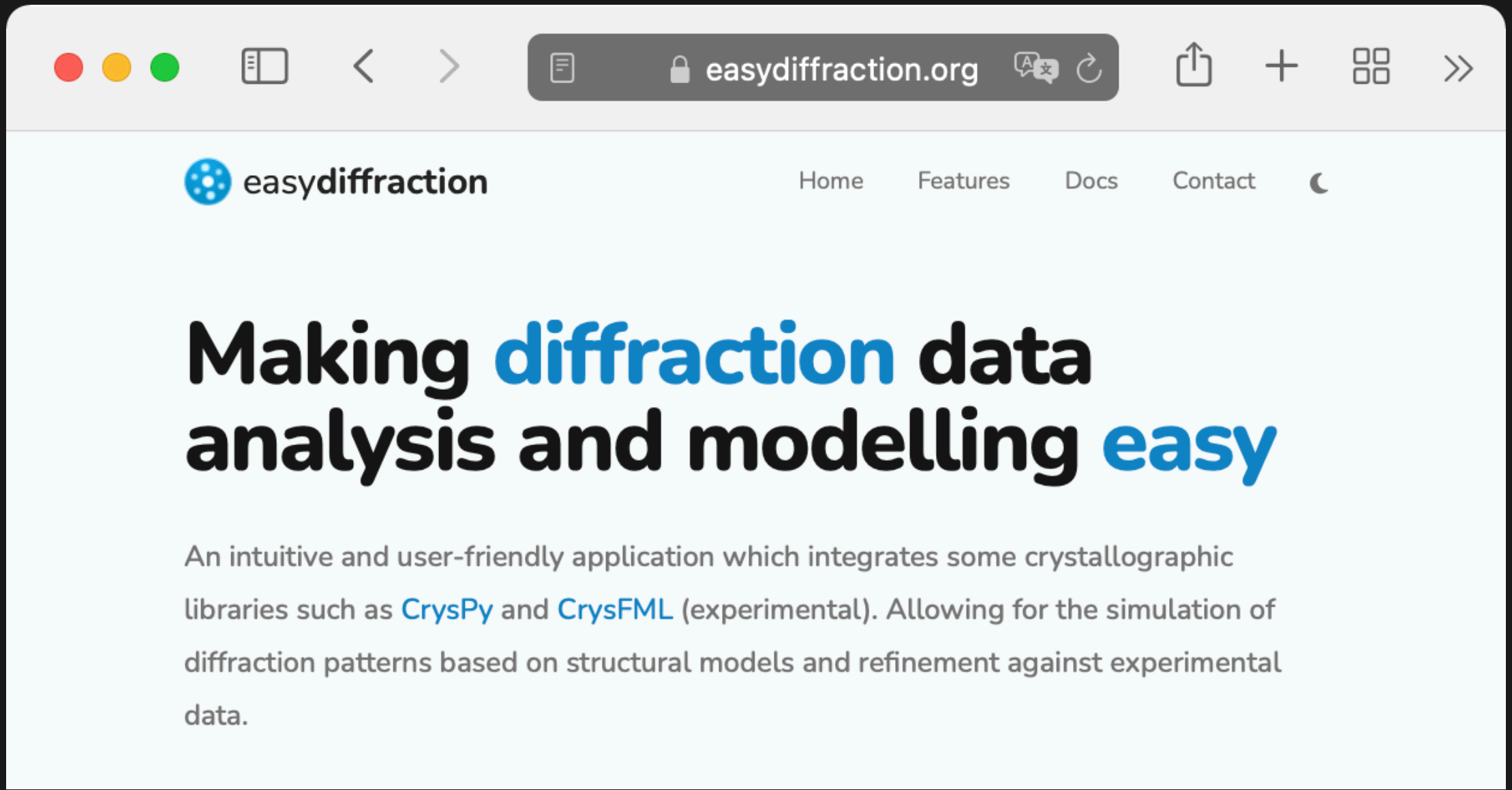


Next steps

Bringing existing features to users.

- **Magnetic structures** · from CrysPy and CrysFML, the biggest gap
- **Polarised neutrons** · from CrysPy
- **Library features in the application** · the library is two years ahead
- **Fixes and extensions** · bugs, usability, small features

Project page · easydiffraction.org



Project page · easydiffraction.org

